PR #213: docs(getting-started): rewrite the install flow around the tt CLI

https://github.com/tenstorrent/tenstorrent.github.io/pull/213
Branch `getting-started-tt-cli` → `main` · author ndalyTT · head `98b6e0aa`

Rendered review copy of the pages this PR changes, built with Sphinx from the PR head (commit `98b6e0a`; a stacked PR already contains its base branch), generated 2026-11-04. Pages render in the site theme with the navigation removed; code blocks wrap instead of scrolling.

Commits (9)

`38864799`	2026-10-23	docs(getting-started): rewrite the install flow around the tt CLI
`18c00e12`	2026-10-23	docs(getting-started): reword the CLI update step and the gated-model…
`2af0032f`	2026-10-23	docs(getting-started): tell the reader what to do when a gated model …
`31c7e7cc`	2026-10-23	docs(getting-started): restore the page title Installing Tenstorrent …
`ba3a0b0b`	2026-10-23	docs(getting-started): refresh the tt model list example
`3dede999`	2026-10-28	docs(getting-started): address review on the tt CLI install guide
`fd115cef`	2026-10-29	docs(getting-started): end the install guide at hardware verification
`e6111397`	2026-10-29	docs(getting-started): note that a fresh TT-QuietBox 2 ships with tt …
`98b6e0aa`	2026-11-04	docs(getting-started): add the Wormhole mesh step and chip-count guid…

Files changed (2)

`core/getting-started/README.md`	+171	-120
`core/getting-started/manual-software-install.md`	+2	-2

Pages in this PDF

1. `getting-started/README.html`
2. `getting-started/manual-software-install.html`

Installing Tenstorrent Software

This guide takes you from a fresh Tenstorrent system to one that is ready to run models. You will install the Tenstorrent CLI, `tt`, use it to update your system to the latest stable Tenstorrent software, and confirm your hardware is visible.

`tt` is the primary interface for operating your Tenstorrent hardware. If you want to develop against Tenstorrent's open-source repositories instead, start with TT-Metalium™, TT-NN™, or TT-Forge™.

Before You Begin

Note

Setting up your new TT-QuietBox® 2? Follow the Setup Guide.

Before you begin, ensure your host machine meets the following prerequisites:

- The host machine has an internet connection. `tt` downloads packages from PyPI and GitHub, the Tenstorrent package repositories, and model weights from the Hugging Face Hub.
- The host machine has a supported operating system installed (see below).
- You have administrator (`sudo`) privileges on the host machine. The driver, firmware tooling, and HugePages configuration are installed as root.
- The system is physically unboxed and set up according to your product's installation manual and safety guidelines.

Supported Operating Systems

Tenstorrent recommends Ubuntu LTS for all Tenstorrent software.

Operating system	Status
Ubuntu 22.04 LTS, Ubuntu 24.04 LTS	Supported
Other Linux distributions	Untested and considered experimental
macOS, Windows, WSL	Not supported. There is no driver, so nothing hardware-related work

BIOS Requirements

The BIOS for your host motherboard should configure the PCIe AER Reporting Mechanism to be set to `OS First`. Tenstorrent's TT-SMI software requires this setting to function properly.

Important

If you are using a TT-QuietBox, you do not have to worry about setting this option. It is set to `OS First` by default.

If you update or reset your BIOS for any reason, you must reconfigure the PCIe AER Reporting Mechanism setting to `OS First` to ensure TT-SMI functions correctly. This setting is typically located in the **Motherboard Information** section of your BIOS.

Install the Tenstorrent CLI Dependencies

The Tenstorrent CLI, `tt`, is a Python application. It needs Python 3.10 or newer with the `venv` module, plus `git` and `curl`. Install them with your system package manager. On Ubuntu:

```
sudo apt update && sudo apt install -y python3 python3-venv python3-pip git curl ca-certificates
```

`tt` must live in its own isolated Python environment so that it can safely update itself later. Tenstorrent recommends uv to manage that environment. Follow the setup instructions at https://docs.astral.sh/uv/ to install `uv`, then open a new terminal or source your shell configuration file (`source ~/.bashrc` for `bash`) so that `uv` is on your `PATH`.

Note

If you already use pipx, it works too. Install `tt` with `pipx install tenstorrent` in the next step instead of the `uv` command. Other installation options are described in the tt-cli developer notes.

Step 1: Install the Tenstorrent CLI

Note

If you're starting with a fresh TT-QuietBox® 2, we automatically install the Tenstorrent CLI for you. To check, run `tt --version` in your terminal. If you get a response, move to Step 2 below.

Install `tt` as a `uv` tool and make sure it is on your `PATH`:

```
uv tool install tenstorrent
uv tool update-shell
```

Open a new terminal or source your shell configuration file again, then confirm `tt` runs:

```
tt --version
```

The first time you run `tt` in a terminal it asks once whether you want to share anonymous usage telemetry. Please consider saying yes. Telemetry tells Tenstorrent which commands people use, how long they take, and which ones fail, and that is what guides what gets fixed and built next. It is strictly opt-in and records no prompts, file paths, hostnames, usernames, or IP addresses. You can see exactly what would be sent before you decide: set `TT_TELEMETRY_LOG_FILE=/tmp/tt-events.jsonl` on any command and read the file, which works without opting in and uploads nothing. You can change your answer at any time with `tt config set telemetry.enabled true` or `false`. The telemetry policy lists everything that is collected.

Run `tt` to see the command groups. The ones used in this guide are `update` and `device`; `model`, `serve`, and `launch` take over once the stack is installed.

Step 2: Install the Software Stack with `tt update`

Tenstorrent's system software stack consists of several components that are regularly tested together to ensure stability: device firmware, the kernel-mode driver (TT-KMD), HugePages configuration, TT-SMI, TT-Flash, and a container runtime for serving models. `tt update` brings your system up to the latest stable, tested combination of those components in one command. On a fresh machine it installs everything. On a machine that already has Tenstorrent software it changes only what is out of date, so you can rerun it at any time.

```
tt update
```

`tt update` installs the Tenstorrent tools into isolated environments under `~/.local/share/tenstorrent/`, where they cannot conflict with your own Python environment, then runs the matching tt-installer release non-interactively to install the driver, firmware, HugePages configuration, and Docker if no container runtime is present.

Required

Using sudo is required, so you must enter your user's password when prompted.

`tt update` also asks you to confirm before it flashes firmware, because flashing resets the devices. Stop any running model workloads first, then answer **y**.

On a TT-QuietBox 2 with a fast connection the whole update takes a few minutes.

Bring `tt` itself up to date

The installer that `tt update` runs also installs the `tt` release that was tested with that set of components, which can be older than the one you installed in Step 1. This page documents the latest release, so before you move on, put `tt` back on it:

```
tt self update
```

Keep both halves current over time: `tt update` for the system software and tools, and `tt self update` for `tt` itself. `tt` checks for a newer release once a day and prints a notice when one is available.

Reboot after the first install

`tt update` never reboots your machine for you. If this is the first time the driver was installed on this machine, or HugePages configuration changed, **reboot now** before continuing:

```
sudo reboot
```

If `tt update` installed Docker, it also adds your user to the `docker` group. That takes effect when you log in again, which the reboot takes care of.

Step 3: Verify Your Hardware

After rebooting, confirm the driver is loaded and that every Tenstorrent device in your system is visible:

```
tt device status
```

You should see one row per chip, with the board type, PCIe bus address, and live telemetry. The **Board** column tells you which generation you have: `p100`, `p150`, and `p300` boards are Blackhole; `n150` and `n300` boards are Wormhole. A few later steps apply to one generation only and say so. Here is an example from a TT-QuietBox 2 with four Blackhole p300 cards:

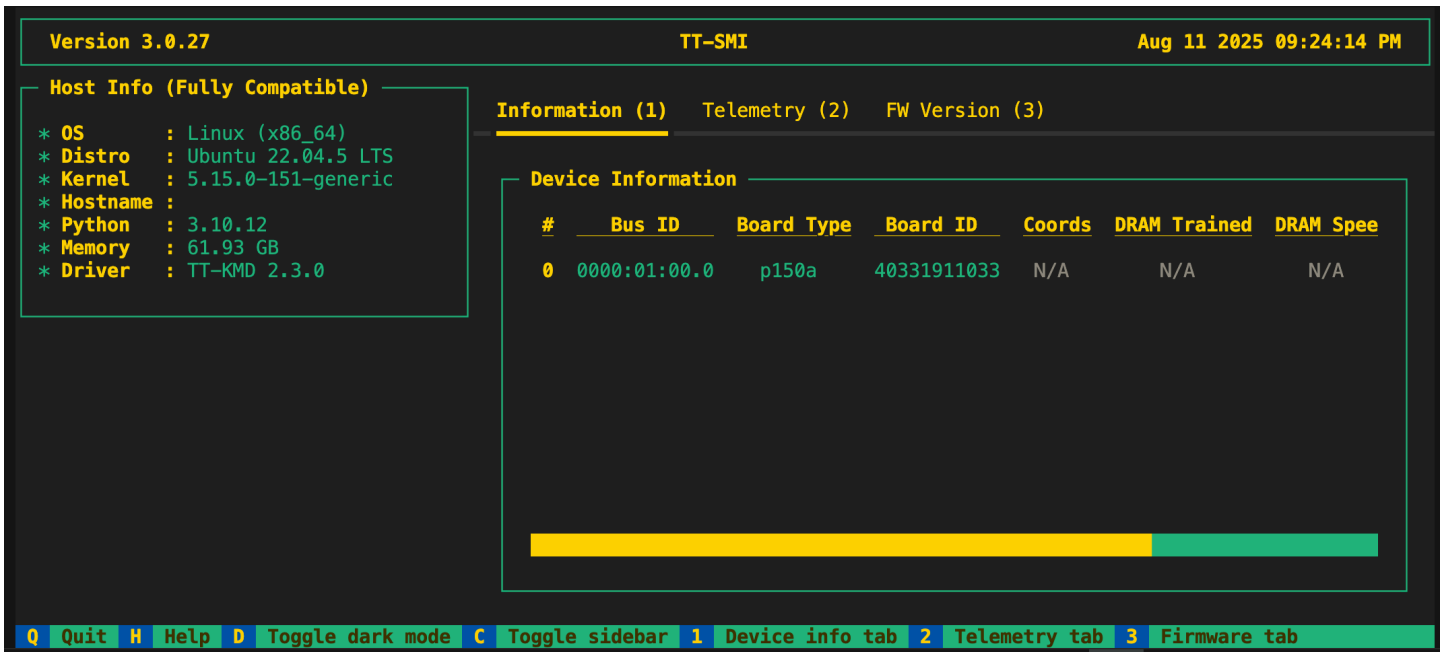
```
                Tenstorrent devices (driver: TT-KMD 2.11.0)
┏━━━┳━━━━━━━┳━━━━━━━━━━━━━┳━━━━━━━━┳━━━━━━━━┳━━━━━━━━━┳━━━━━━━━━┳━━━━━━┓
┃ # ┃ Board ┃ Bus ID      ┃ Temp   ┃ Power  ┃ AICLK   ┃ Voltage ┃ DRAM ┃
┡━━━╇━━━━━━━╇━━━━━━━━━━━━━╇━━━━━━━━╇━━━━━━━━╇━━━━━━━━━╇━━━━━━━━━╇━━━━━━┩
│ 0 │ p300c │ 0000:01:00.0 │ 43.9 °C │ 31.0 W │ 800.0 MHz │ 0.71 V  │ True │
│ 1 │ p300c │ 0000:02:00.0 │ 44.4 °C │ 17.0 W │ 800.0 MHz │ 0.71 V  │ True │
│ 2 │ p300c │ 0000:03:00.0 │ 43.6 °C │ 16.0 W │ 800.0 MHz │ 0.72 V  │ True │
│ 3 │ p300c │ 0000:04:00.0 │ 45.1 °C │ 16.0 W │ 800.0 MHz │ 0.72 V  │ True │
└───┴───────┴──────────────┴────────┴────────┴─────────┴─────────┴──────┘
```

A Wormhole n300 card carries two chips and shows two rows, `n300 L` with the card's PCIe bus address and `n300 R` with `N/A`, because the second chip is reached through the first. A TT-LoudBox or Wormhole TT-QuietBox with four n300 cards therefore shows eight rows:

```
                Tenstorrent devices (driver: TT-KMD 2.11.0)
┏━━━┳━━━━━━━┳━━━━━━━━━━━━━┳━━━━━━━━┳━━━━━━━━┳━━━━━━━━━━┳━━━━━━━━━┳━━━━━━━┓
┃ # ┃ Board ┃ Bus ID      ┃ Temp   ┃ Power  ┃ AICLK    ┃ Voltage ┃ DRAM  ┃
┡━━━╇━━━━━━━╇━━━━━━━━━━━━━╇━━━━━━━━╇━━━━━━━━╇━━━━━━━━━━╇━━━━━━━━━╇━━━━━━━┩
│ 0 │ n300 L │ 0000:31:00.0 │ 45.9 °C │ 28.0 W │ 1000.0 MHz │ 0.89 V  │ True  │
│ 1 │ n300 L │ 0000:4b:00.0 │ 44.8 °C │ 27.0 W │ 1000.0 MHz │ 0.88 V  │ True  │
│ 2 │ n300 L │ 0000:b1:00.0 │ 44.4 °C │ 29.0 W │ 1000.0 MHz │ 0.9 V   │ True  │
│ 3 │ n300 L │ 0000:ca:00.0 │ 43.1 °C │ 29.0 W │ 1000.0 MHz │ 0.9 V   │ True  │
│ 4 │ n300 R │ N/A          │ 37.5 °C │ 24.0 W │ 1000.0 MHz │ 0.87 V  │ True  │
│ 5 │ n300 R │ N/A          │ 36.4 °C │ 20.0 W │ 1000.0 MHz │ 0.91 V  │ True  │
│ 6 │ n300 R │ N/A          │ 37.4 °C │ 24.0 W │ 1000.0 MHz │ 0.91 V  │ True  │
│ 7 │ n300 R │ N/A          │ 36.1 °C │ 23.0 W │ 1000.0 MHz │ 0.9 V   │ True  │
└───┴────────┴──────────────┴─────────┴────────┴────────────┴─────────┴───────┘
```

The number of rows should match the number of Tenstorrent chips installed in your system: one per Blackhole card, two per n300 card. For more detail on a device, such as its firmware versions and board ID, run `tt device info`. For a live, interactive view, run `tt smi`, which hands your terminal to the TT-SMI interface. Press **Q** to leave it.

Here is an example of the TT-SMI interface for a system containing a single p150a device:



Important

If the number of listed devices does not match what you expect, see Troubleshooting below. If the problem persists, contact support and we will assist you.

Congratulations, your system software is installed and your hardware is ready to use.

Step 4 (Wormhole only): Configure the Mesh Topology

Skip this step on Blackhole systems and on single-card Wormhole systems.

A Wormhole TT-LoudBox or TT-QuietBox connects its n300 cards to each other over Ethernet, and the models in the catalog expect those links to form a mesh. `tt update` installs the driver and firmware but does not configure the mesh, and a system can arrive, or come back from a firmware update, with the chips **Isolated**. Check first:

```
tt-topology -ls
```

If it reports `Mesh`, continue to the next page. If it reports `Isolated`, or the command is not found, run the script below once. It installs TT-Topology into a temporary environment, resets every chip, and writes the mesh configuration. It needs no `sudo`, takes about two minutes, and ends with "Completed reset on 8 chips" on a four-card system.

```
TMP_DIR=$(mktemp -d); (trap 'echo "---"; echo "Cleaning up..."; if type deactivate &>/dev/null; then deactivate; fi; echo "Removing temporary directory: $TMP_DIR"; rm -rf "$TMP_DIR"; cd; echo "Cleanup complete."' EXIT; trap 'echo -e "\033[0;31m!!! ERROR: Failed to configure mesh topology\033[0m"' ERR; set -e; cd "$TMP_DIR"; echo "Working in temporary directory: $TMP_DIR"; echo "---"; echo "Creating Python virtual environment..."; python3 -m venv tt-topology-venv; source tt-topology-venv/bin/activate; echo "Virtual environment activated."; echo "---"; echo "Installing tt-topology from git..."; pip install --quiet git+https://github.com/tenstorrent/tt-topology.git; echo "tt-topology installed."; echo "---"; echo "Running tt-topology command. This may take a moment..."; tt-topology -l mesh; echo "---"; echo "Script finished successfully.";)
```

Run `tt-topology -ls` again and confirm `Mesh`. Serving a model without this step stops at system validation with `ModelSpec requires SystemTopology.MESH, detected ISOLATED`.

Troubleshooting

`tt device status` reports no devices

If `tt device status` exits with a message that the driver may not be installed or no card is seated:

1. If you have not rebooted since `tt update` installed the driver, reboot now.
2. Run `lspci -d 1e52:` to check whether the card has enumerated on the PCIe bus. If it returns nothing, the host cannot see the card at all. Each Blackhole card and each n150 card appears once; an n300 card also appears once, because its second chip has no PCIe address of its own.
3. For Blackhole PCIe products such as p100 and p150, verify the card is being supplied power by checking that the fan spins and the green power LED illuminates when the system boots. If the light is on, try resetting the card with `tt device reset`.
4. Confirm the BIOS setting described in BIOS Requirements.

Docker commands are refused with a permission error

`tt` talks to Docker as your user, never through `sudo`. If `tt update` just installed Docker, log out and back in (or reboot) so that your membership in the `docker` group takes effect. If Docker was already installed before you ran `tt update`, the installer leaves it as it found it and does not add you to the group. Add yourself, then log out and back in:

```
sudo usermod -aG docker $USER
```

You can check with `docker run hello-world`.

Getting help

`tt report bundle` collects a redacted support bundle with your environment details, a TT-SMI snapshot, and recent logs, and opens a prefilled email to Tenstorrent support with it attached. `tt report issue` opens a prefilled GitHub issue on the tt-cli repository.

Alternative Installation Methods

`tt update` is the recommended way to install the system software, and it runs tt-installer for you with the tested versions. If you prefer to drive tt-installer interactively yourself and choose each of its options, you can run it directly:

```
/bin/bash -c "$(curl -fsSL https://tenstorrent.ai/install.sh)"
```

For advanced users or developers who prefer to install and configure each component by hand, refer to the manual software installation guide.

What's Next

With the software stack installed, `tt` takes you the rest of the way to a running model. `tt model list` shows the models validated for the hardware it detected, and `tt model info <model>` reports the disk space, host memory, and validation status of each one. Catalog models download their weights from the Hugging Face Hub, so log in once with `tt model login`; gated models such as Llama and Gemma also require accepting the license on the model's Hub page first. `tt serve <model>` then installs TT-Inference-Server, downloads the weights, and starts an OpenAI-compatible endpoint on port 20000, and `tt model curl "<prompt>"` sends it a first request. The first serve of a model downloads and compiles a lot, so expect it to take a while; later serves are much faster.

An endpoint is the foundation; `tt launch` puts an application on top of it. `tt launch openwebui` starts a chat interface in your browser, and `tt launch list` shows the other web UIs and terminal coding agents available on your machine. If you prefer a point-and-click experience, `tt serve <model> --studio` starts the model inside TT-Studio, a web application that serves models and gives you a browser interface for them in one place. Run `tt` at any time to see the command groups, and `tt <command> --help` for the options of each.

Where to Go Next

You now have the full stack installed. From here, pick the part of the stack that matches what you want to build:

- **Deploy LLMs in production** with TT-Inference-Server, the serving engine behind `tt serve`. The tt-inference-server repository is the authoritative source for which models are validated on your hardware.
- **Monitor your hardware** with `tt smi` for a status snapshot, or TT-Toplike for a live view of power, temperature, and core activity while a model serves requests.
- **Use a web interface** with TT-Studio, started through `tt serve <model> --studio`.
- **Understand the software stack**, including how TT-Forge™, TT-NN™, and TT-Metalium™ relate to each other, in the software stack overview.
- **Compile your own PyTorch, JAX, or ONNX models** with TT-Forge™.
- **Run and build models in Python** with TT-NN™, or **write custom kernels** with TT-Metalium™. `tt update` also installed a `tt-metalium` command that opens a shell inside the TT-Metalium container, with your home directory mounted and TT-NN ready to import; the image, about 4.5 GB, is pulled the first time you run it. Start with the Metalium guide to learn about the architecture.
- **Learn by building** with the hands-on lessons in the TT VS Code Toolkit.

Need Additional Support?

If you encounter any issues, or have a question that isn't covered in the documentation, please raise a support request. Our team will review your request and provide assistance.

Manual Installation

Important! At this time, the recommended method for installing Tenstorrent software is tt-installer. If you would prefer to install software dependencies and configure your system manually, proceed with the instructions below. When you are done, return to the Quickstart Guide for next steps.

This Starting Guide will reference each software utility where the latest version is available. However, each SDK will have its own compatibility matrix associated with each release. It is strongly recommended to consult each SDK's release compatibility matrix to ensure you are installing the correct versions of the system software packages.

Step 1: Install Software Dependencies

Install git, wget, pip, and DKMS (Dynamic Kernel Module Support) by running the following command in your terminal according to your Linux distribution:

Linux Distro	Installation Command
Debian, Ubuntu	`sudo apt update && sudo apt install -y wget git python3-pip dkms cargo`
Fedora	`sudo dnf check-update && sudo dnf install -y wget git python3-pip dkms cargo`
Enterprise Linux based	`sudo dnf install -y epel-release && sudo dnf check-update && sudo dnf install -y wget git python3-pip dkms cargo`

***NOTE**: Installation on non-Ubuntu distributions should be considered experimental at this time.*

Step 2: Install the Kernel-Mode Driver (TT-KMD)

Install the driver (**TT-KMD**) by running these commands in the terminal:

```
git clone https://github.com/tenstorrent/tt-kmd.git
cd tt-kmd
sudo dkms add .
sudo dkms install tenstorrent/1.34
sudo modprobe tenstorrent
```

Step 3: Update Device Firmware (TT-Flash / TT-System-Firmware)

Use the **TT-Flash** utility to flash the latest **TT-System-Firmware** onto your Tenstorrent device.

Install TT-Flash

To install **TT-Flash**, run this command in the terminal:

```
pip install git+https://github.com/tenstorrent/tt-flash.git
```

***NOTE:** If you are not using a Python virtual environment (venv), you may see an error `externally-managed-environment` when installing via `pip`. To resolve this, create and/or activate a venv or use a tool like pipx.*

Update Device Firmware

⚠️ NOTE: Blackhole devices require firmware version 18.3.0 or newer.

To update Tenstorrent device firmware using TT-Flash, run these commands in the terminal:

```
wget https://github.com/tenstorrent/tt-system-firmware/releases/download/v18.4.0/fw_pack-18.4.0.fwbundle
tt-flash --fw-tar fw_pack-18.4.0.fwbundle
```

If this process worked, reboot the system and go to the next section.

If running that command results in an error that says the firmware is too old, enter the following command:

```
tt-flash --fw-tar fw_pack-18.4.0.fwbundle --force
```

Then reboot the system.

Step 4: Set Up HugePages

HugePages lets your system allocate dedicated memory to accelerate communication with Tenstorrent devices. Set up HugePages by running these commands in the terminal:

```
# Install `.deb`
wget https://github.com/tenstorrent/tt-system-tools/releases/download/v1.3.1/tenstorrent-
tools_1.3.1_all.deb
sudo dpkg -i tenstorrent-tools_1.3.1_all.deb

# Start Services
sudo systemctl enable --now tenstorrent-hugepages.service
sudo systemctl enable --now 'dev-hugepages\x2d1G.mount'

# System Reboot
sudo reboot
```

***NOTE:** This is a temporary solution for configuring hugepages. If the above fails, please check the latest available release from TT-System-Tools.**

Step 5: Multi-Card Configuration (TT-Topology)

NOTE: Multi-card Wormhole systems need a mesh topology before the catalog models will serve. Run `tt-topology -ls` first; if it reports `Mesh` you can skip this step. TT-Topology is provided as-is.

If you are running on a multi-card Wormhole system such as TT-LoudBox or TT-QuietBox, install the Tenstorrent Topology utility (**TT-Topology**) and configure a mesh topology by running these commands in the terminal:

```
pip install git+https://github.com/tenstorrent/tt-topology
tt-topology -l mesh
```

Step 6: Install the System Management Interface (TT-SMI)

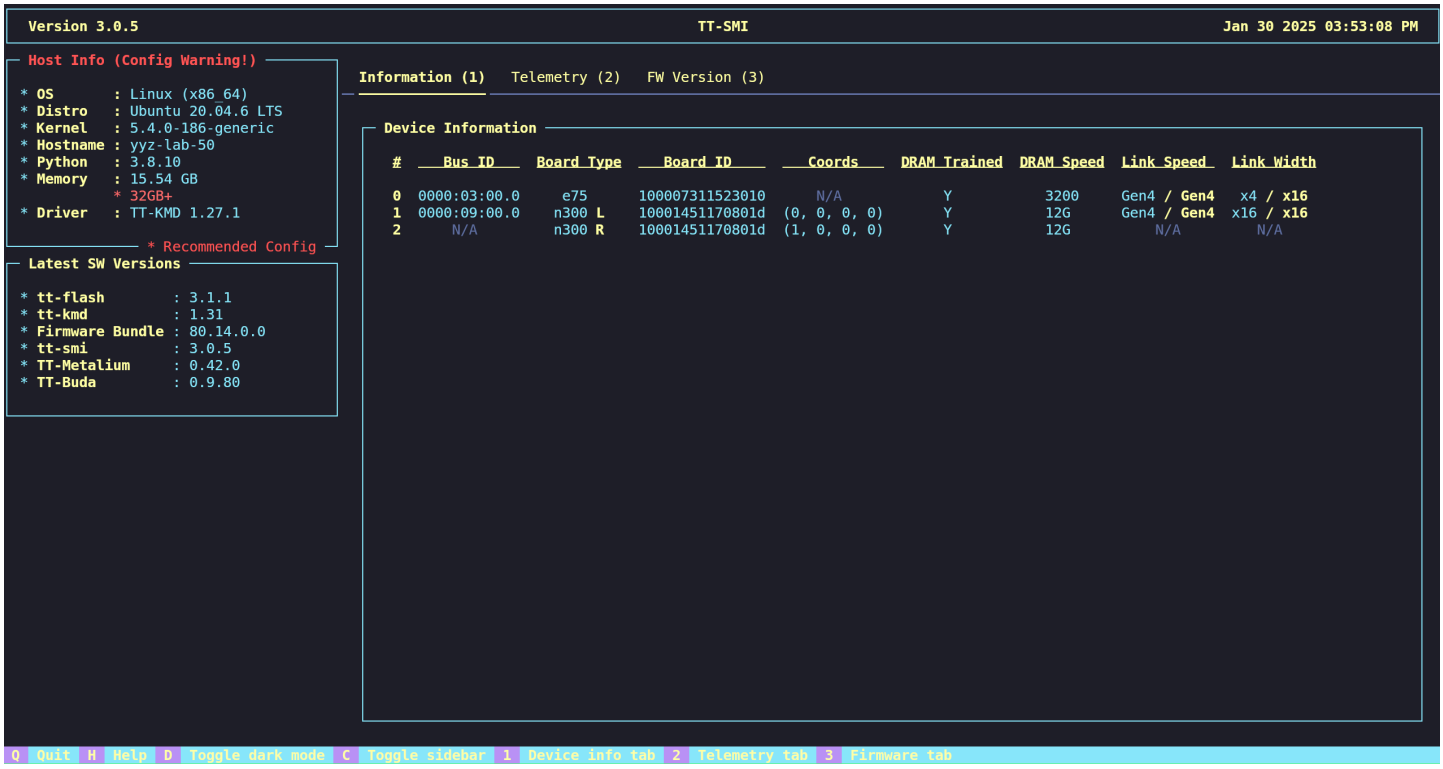
Install the Tenstorrent Software Management Interface (**TT-SMI**) by entering this command in the terminal:

```
pip install git+https://github.com/tenstorrent/tt-smi
```

Step 7: Verify System Configuration and Test TT-SMI

Once your hardware and system software are installed, verify that your system has been configured properly by running the `tt-smi` utility.

You should see an interface like this one:



Within TT-SMI, you can see device information, telemetry, and firmware. If TT-SMI runs without errors, congratulations! You're ready to use your Tenstorrent device. Take note of any host compatibility warnings.

If you are using a TT-QuietBox and have reboot issues while using TT-SMI, and you've recently updated/reset the BIOS, you will need to update a setting. In the BIOS, find the following setting:

Advanced -> AMD CBS -> NBIO Common Options -> PCIe Aer Reporting Mechanism

And change the setting to **OS First**.